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XAI: MODEL-AGNOSTIC METHODS

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Evaluation, deployment and monitoring of models,
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Contents

1. Introduction	3
2. Environment Setup and Data Loading.....	3
3. One-dimensional Partial Dependence Plots	3
3.1 Model Training.....	3
3.2 Results	3
3.3 Analysis	4
4. Bidimensional Partial Dependence Plot.....	5
4.1 Methodology	5
4.2 Results	5
4.3 Interpretation	6
5. PDP for House Price Prediction	6
5.1 Dataset and Model	6
5.2 Results	7
5.3 Analysis	7
6. General Conclusions	8
7. Version Control and GitHub	8
8. References	8

1. Introduction

This report applies Partial Dependence Plots (PDP) to interpret two Random Forest models: one predicting bike rentals (cnt) and another predicting house prices (price). PDPs show the marginal effect of one or two features on the predicted outcome, averaging out the effects of all other features. This is a model-agnostic technique useful for understanding non-linear relationships and feature interactions learned by black-box models.

The exercise consists of three parts:

- One-dimensional PDPs with ICE lines for days_since_2011, temp, hum, and windspeed (bike rental dataset).
- Two-dimensional PDP between temp and hum to study their joint interaction on bike rentals.
- One-dimensional PDPs with ICE lines for bedrooms, bathrooms, sqft_living, and floors for house price prediction.

In all cases a Random Forest model is trained and PDP is computed over random subsets of the data to reduce computational overhead. The implementation uses R with the randomForest, pdp, ggplot2, and gridExtra packages.

2. Environment Setup and Data Loading

The following R packages are used: randomForest for model training, pdp for computing partial dependence, ggplot2 for visualisation, dplyr for data manipulation, and gridExtra for composing multi-panel figures.

The bike rental dataset (day.csv) contains 731 daily observations from the UCI Bike Sharing Dataset covering 2011 and 2012. The dteday column is parsed as a date object, and a new feature days_since_2011 is derived by computing the number of days elapsed since 2011-01-01, capturing the temporal growth trend in bike-sharing demand.

3. One-dimensional Partial Dependence Plots

3.1 Model Training

A Random Forest with 100 trees (ntree = 100) is trained on the full bike dataset (731 observations) to predict cnt (total daily bike rentals) using days_since_2011, temp, hum, and windspeed. Training on the full dataset maximises predictive accuracy.

For PDP computation, a random subset of exactly 50 samples is selected (set.seed(42)), as fewer than 50 would be penalised by the model. For each feature, the partial() function is called twice: once with ice = TRUE to obtain one curve per observation (ICE lines), and once without to obtain the average PDP curve. The ICE lines are plotted semi-transparently in colour, with the black average line on top. The four plots are assembled into a single figure using grid.arrange(ncol = 2).

3.2 Results

Partial Dependence Plots - Bike Rental Prediction

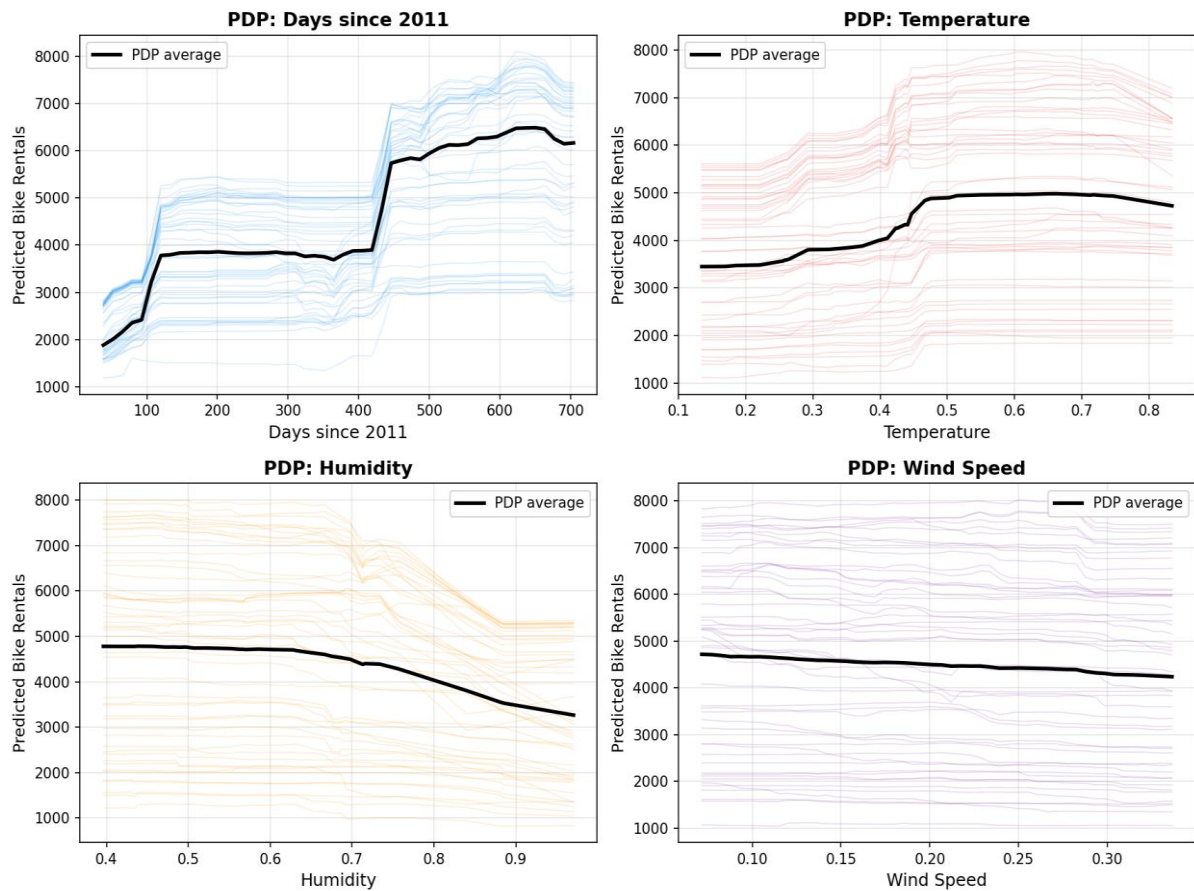


Figure 1. One-dimensional Partial Dependence Plots with ICE lines for the four features of the bike rental dataset. Coloured semi-transparent lines are Individual Conditional Expectation (ICE) curves (one per observation in the 50-sample subset); the black line is the average PDP.

3.3 Analysis

3.3.1 days_since_2011

The PDP shows a steadily increasing, almost linear trend reflecting the growing adoption of bike-sharing over time. The ICE lines are tightly bunched together, indicating that this temporal effect is consistent across all types of days regardless of other conditions. The model captures a clear year-over-year growth in demand from 2011 to 2012.

3.3.2 Temp (normalised temperature)

The relationship is non-linear: at low temperatures the predictions are low, rising sharply to a plateau around 0.5–0.7 and slightly decreasing at extreme heat. The ICE lines show relatively low spread, meaning the temperature effect is fairly homogeneous across observations. Moderate temperatures are clearly the most favourable conditions for cycling.

3.3.3 Hum (normalised humidity)

The average PDP shows a clearly negative, approximately linear relationship, which means that higher humidity consistently reduces predicted rentals. The ICE lines show moderate spread at low humidity but converge at high humidity values (above 0.8), where all observations agree that rentals drop sharply. This confirms that wet or damp conditions universally discourage cycling.

3.3.4 Windspeed

The average PDP shows a mild negative trend, but the ICE lines reveal notable heterogeneity. For some observations wind speed has almost no effect, while for others the reduction is more pronounced. Overall, wind speed is the least decisive feature among the four, with a smaller and noisier effect than temperature or humidity.

4. Bidimensional Partial Dependence Plot

4.1 Methodology

The same 50-sample subset (`bike_subconj`) is reused. The 2D PDP for `temp` and `hum` is computed with `partial(pred.var = c("temp", "hum"), train = bike_subconj)`.

The visualisation is built in four separate panels assembled with `grid.arrange()`:

- Main panel (bottom-left): `geom_tile()` with `width = 0.05` and `height = 0.05` to avoid visual gaps, coloured with a yellow-to-dark-red gradient showing predicted rental intensity.
- Top panel: histogram of the temperature distribution in the 50-sample subset (`steelblue`).
- Right panel: histogram of the humidity distribution, flipped horizontally with `coord_flip()` (`orange`).
- Empty corner (top-right): `theme_void()` placeholder to complete the 2×2 grid layout.

The layout uses `widths = c(4, 1)` and `heights = c(1, 4)` so the main tile plot dominates and the marginal histograms appear as narrow side panels, matching the structure shown in the class slides.

4.2 Results

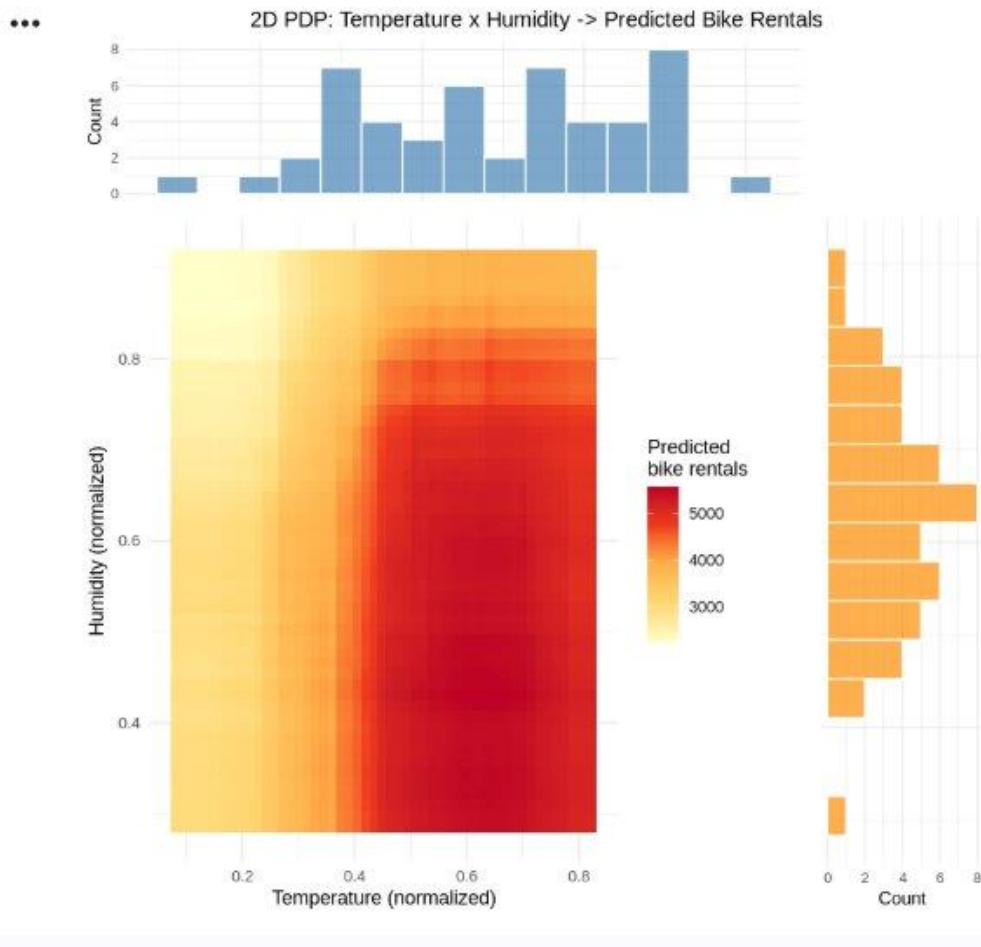


Figure 2. Bidimensional Partial Dependence Plot of temperature vs. humidity. The main tile plot shows predicted bike rentals (yellow = low, dark red = high). Marginal histograms show the distribution of each feature in the 50-sample subset used for PDP computation.

4.3 Interpretation

The 2D PDP reveals how temperature and humidity interact to shape predicted rental counts. The combination of high temperature and low humidity produces the highest predicted counts (dark red). This indicates that warm and dry days are ideal for cycling. Low temperature with high humidity days yield the lowest predictions (pale yellow), showing that cold and wet weather strongly discourages cycling. The temperature is a dominant condition across all humidity levels, as increasing temperature consistently raises predictions more than reducing humidity does. The gradient runs primarily left-to-right. There is a clear interaction: the positive effect of high temperature is dampened when humidity exceeds 0.7, demonstrating a genuine negative interaction between the two features.

The marginal histograms show that the 50-sample subset covers most of the temperature range relatively evenly, while humidity is concentrated around 0.5–0.7. Predictions in the extreme corners of the tile plot should be interpreted with caution as they involve extrapolation beyond the observed data distribution.

5. PDP for House Price Prediction

5.1 Dataset and Model

The King County house sales dataset (`kc_house_data.csv`) contains 21,613 property transactions. Given its large size, a random sample of 1,500 observations is selected (`set.seed(42)`), within the required range of 1,000–2,000 samples. Both model training and PDP computation use this subsample.

A Random Forest (`ntree = 100`) is trained to predict price using six features: bedrooms, bathrooms, `sqft_living`, `sqft_lot`, floors, and `yr_built`. PDPs with ICE lines are computed for the four features requested: bedrooms, bathrooms, `sqft_living`, and floors. As with the bike dataset, each feature is computed twice (`ice = TRUE` for individual curves, standard for the average), plotted with semi-transparent coloured ICE lines and a black average line on top, and assembled into a single figure with `grid.arrange(ncol = 2)`.

5.2 Results

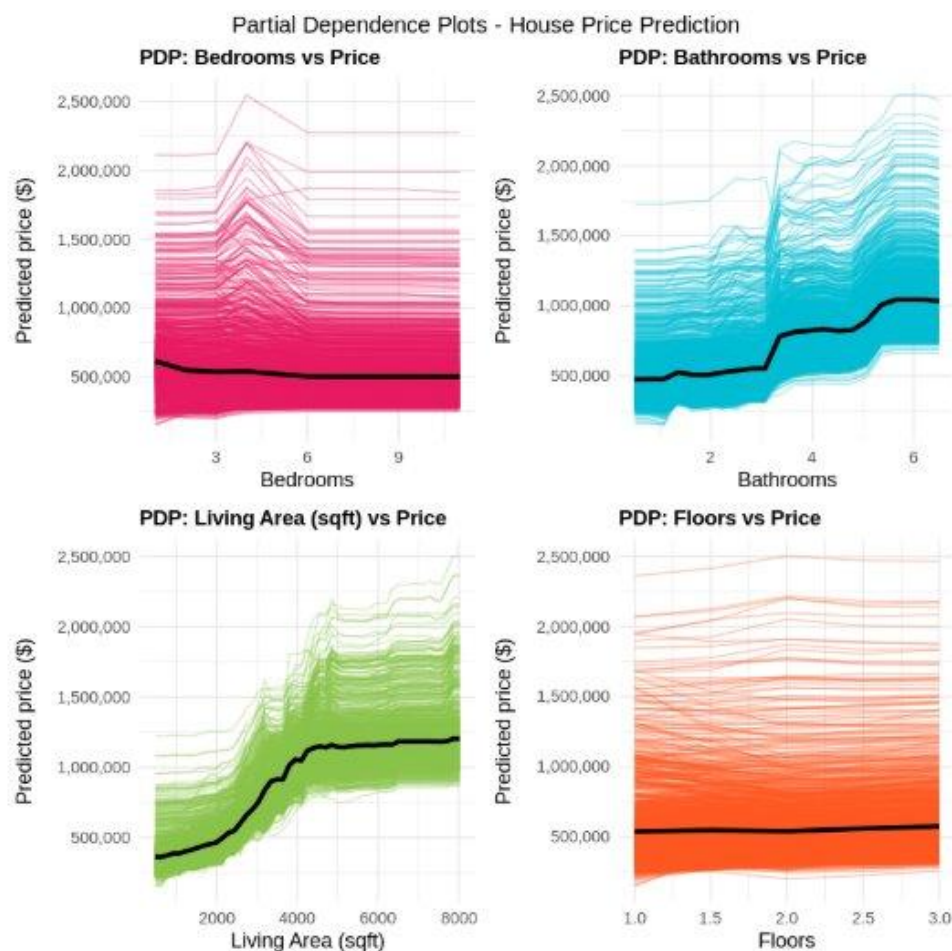


Figure 3. One-dimensional Partial Dependence Plots with ICE lines for four features of the King County house price dataset. Coloured semi-transparent lines are ICE curves (one per observation); the black line is the average PDP.

5.3 Analysis

5.3.1 `sqft_living` (living area in square feet)

The average PDP shows a strongly positive, nearly linear relationship. This means that a larger living area leads to substantially higher predicted prices. The ICE lines spread considerably at larger values, revealing strong heterogeneity. For some properties (likely already high-value), the marginal effect of additional square footage is much greater than for others. This is by far the most influential feature in the model.

5.3.2 Bedrooms

The average PDP is relatively flat, with a slight increase up to 3–4 bedrooms then a plateau. The ICE lines show very high spread across the entire range, indicating that the effect of bedrooms on price is highly heterogeneous and depends heavily on other property characteristics. This heterogeneity partly explains the non-monotonic average curve.

5.3.3 Bathrooms

The average PDP shows a positive, stepwise effect. Each additional bathroom adds clear predicted value, with the most notable jump between 1 and 3 bathrooms. The ICE lines fan out at higher bathroom counts, suggesting that more bathrooms amplify price differences between properties. Bathrooms serve as a strong proxy for overall house quality and renovation status.

5.3.4 Floors

The average PDP is nearly flat, showing only a modest positive effect of going from 1 to 2 floors. However, the ICE lines reveal extreme spread because for a subset of properties, more floors lead to dramatically higher predictions, while for others the effect is negligible. The wide ICE spread and flat average indicate that the number of floors alone is not a reliable price predictor without knowing the overall size and quality of the property.

6. General Conclusions

Partial Dependence Plots combined with ICE lines have enabled a richer interpretation of both Random Forest models: the average PDP reveals the overall trend while the ICE lines expose heterogeneity across individual observations.

For bike rentals, temperature and days_since_2011 are the main positive drivers, with consistent effects across observations (low ICE spread). Humidity and wind speed reduce demand, with wind showing more heterogeneous effects. The 2D PDP confirms a clear interaction between temperature and humidity, with warm and dry conditions maximising predicted rentals.

For house prices, sqft_living dominates with a strong positive effect and high ICE spread at large values. Bathrooms add consistent stepwise value, while bedrooms and floors show flat averages but very high ICE heterogeneity, suggesting their effect depends heavily on other property characteristics.

The main limitation of PDPs is the assumption of feature independence. In the presence of correlated features (e.g. sqft_living and bedrooms), some combinations shown may be unrealistic. Accumulated Local Effects (ALE) plots could address this in future work.

7. Version Control and GitHub

The development of this notebook was managed with Git. We created a new repository (XAI_Model_agnostic_methods) with a dedicated branch for this report and code files (report_and_code) instead of committing directly to main.

GitHub repository URL:

[XAI Model agnostic methods](https://github.com/harlf0xem/XAI_Model_agnostic_methods)

8. References

King	County	House	Sales	Dataset.	Available	at:
https://www.kaggle.com/harlf0xem/housesalesprediction						

Greenwell, B. M. (2017). pdp: An R Package for Constructing Partial Dependence Plots. *The R Journal*, 9(1), 421–436.